

Sankaraleengam (Sankar) Alagapan

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RESEARCH INTERESTS	Brain stimulation and data science with a focus on understanding brain dynamics in cognition and psychiatric disorders	
EDUCATION	Ph.D. Biomedical Engineering University of Florida, Gainesville, FL	2014
	M.S. Biomedical Engineering University of Florida, Gainesville, FL	2008
	B.E. Electrical and Electronics Engineering Anna University, Chennai, India	2007
RESEARCH EXPERIENCE	Research Scientist II Georgia Institute of Technology	2022 – Present
	Research Engineer II Georgia Institute of Technology	2018 – 2022
	Postdoctoral Research Associate University of North Carolina at Chapel Hill	2014 – 2018
	Graduate Research Assistant University of Florida	2008 – 2014
HONORS & AWARDS	Georgia Institute of Technology Georgia CTSA KL2 Scholar Best Faculty Paper Award – Sigma Xi Roger. P Webb Outstanding Non-Tenure Faculty	2023 2024 2024
	University of North Carolina at Chapel Hill Helen Lyng White Postdoctoral Fellowship Office of Postdoctoral Affairs Travel Award	2016 2016
	University of Florida Office of Research Travel Award Graduate Student Council Travel Award College of Engineering Achievement Award	2010, 2013 2011 2007
RESEARCH SUPPORT	A naturalistic multimodal platform for capturing brain-body interactions in people during physical effort-based decision making NIH NIMH R61MH138966	2025 - 2028

Georgia Clinical and Translational Science Alliance 2023 – 2025
 NIH NCATS KLTR002381
 Role: KL2 Scholar
 Electrophysiological Biomarkers to Optimize DBS for Depression 2019 - 2024
 NIH NINDS UH3 NS103550
 Role: Postdoc/ Team Lead

PUBLICATIONS

[\(Google Scholar\)](#)
Journal Articles

- Chen, Y., Mudrik, N., Johnsen, K.A., **Alagapan, S.**, Charles, A.S. and Rozell, C., (2025). Probabilistic decomposed linear dynamical systems for robust discovery of latent neural dynamics. *Advances in Neural Information Processing Systems*, 37, p 104443-104470.
- Xu E., Pitts S., Dahill-Fuchel J., Scherrer, S., Nauvel T., ..., **Alagapan S.**, ..., Waters A.C., (2024). Neural Interoceptive Processing is Modulated by Deep Brain Stimulation for Treatment Resistant Depression. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, p S2451-9022.
- Van Rheede J.*, **Alagapan S.***, Denison T., Riva Posse P., Rozell C., Mayberg H., Waters A.C., Sharrott A., Cortical signatures of sleep are altered following effective deep brain stimulation for depression. *Translational Psychiatry* 14:1
- Alagapan S.**, Heisig S., Choi K., Riva Posse P., Crowell A., ..., Mayberg H. M., and Rozell, C. (2023) Cingulate dynamics track depression recovery with deep brain stimulation *Nature* 622, pages 130–138
- Smith E.E., Choi K.S., Veerakumar A., Obatusin M., Howell B., Smith A.H., Crowell A., Riva Posse P., **Alagapan S.**, Rozell C.J., Mayberg H.S., and Waters A.C. (2022) Time-frequency signatures evoked by single-pulse deep brain stimulation to the subcallosal cingulate. *Frontiers in Human Neuroscience*, 16:939258
- Wong, J. K., Deuschl, G., Wolke, R., Bergman, H., Muthuraman, M., Groppa, S., ... **Alagapan, S.**, ... and Okun, M. S. (2022). Proceedings of the Ninth Annual Deep Brain Stimulation Think Tank: Advances in Cutting Edge Technologies, Artificial Intelligence, Neuromodulation, Neuroethics, Pain, Interventional Psychiatry, Epilepsy, and Traumatic Brain Injury. *Frontiers in Human Neuroscience*, 16:813387
- Alagapan S.**, Riddle, J., Huang, W.A., Hadar, E., Shin H.W., Fröhlich., (2019) Network-targeted, multi-site direct cortical stimulation enhances working memory by modulating phase lag of low frequency oscillations. *Cell Reports* 29 (9), 2590-2598
- McPherson, T. †, Berger, D.S., **Alagapan, S.**, and Fröhlich, F. (2019) Active and passive music therapy interventions differentially modulate sympathetic autonomic nervous system activity. *Journal of Music Therapy* 56 (3), 240-264
- Alexander, M.L.*, **Alagapan, S.***, Lugo, C.E., Mellin, J.M., Lustenberger, C., Rubinow, D.R., and Fröhlich, F. (2019) Double-blind, randomized pilot clinical trial targeting alpha oscillations with transcranial alternating current stimulation (tACS) for the treatment of major depressive disorder (MDD). *Translational Psychiatry* 9:106
- Alagapan, S.**, Fröhlich, F., and Wu, H-T., Diffusion geometry approach to efficiently remove electrical stimulation artifacts in intracranial electroencephalography. (2018) *Journal of Neural Engineering* 16, 036010

- Sheffield, A. †, Ahn, S., **Alagapan, S.**, and Fröhlich, F. (2018) Investigating Effects of Neural Oscillations by Lateral Transcranial Static Magnetic Field Stimulation on the Prefrontal Cortex: A crossover, double-blind, sham-controlled pilot study. *European Journal of Neuroscience* 00:1–13.
- Ahn S., Mellin, J.M., **Alagapan, S.**, Lugo, C.E., Alexander, M.L., Gilmore, J.H., Jarskog, L.F., and Fröhlich, F. (2018) Targeting impaired neural oscillations in patients with schizophrenia by transcranial alternating current stimulation: A randomized, double-blind, placebo-controlled pilot study. *NeuroImage* 186, 126-136
- Alagapan, S.**, Lustenberger, C., Hadar, E., Shin, H.W., and Fröhlich, F. (2018) Low-frequency direct cortical stimulation of left superior frontal gyrus enhances working memory performance. *NeuroImage* 184, 697-706
- McPherson, T. †, Berger, D.S., **Alagapan, S.**, and Fröhlich, F. (2018) Intrinsic Rhythmicity Predicts Synchronization-Continuation Entrainment Performance. *Scientific Reports* 8, 11782
- Mellin, J.M., **Alagapan, S.**, Lustenberger, C., Lugo, C.E., Alexander, M.L., Gilmore, J.H., Jarskog, L.F., and Fröhlich, F. (2018). Randomized Trial of Transcranial Alternating Current Stimulation for Treatment of Auditory Hallucinations in Schizophrenia. *European Psychiatry* 51, 25-33.
- Lustenberger, C., Patel, Y.A., **Alagapan, S.**, Page, J.M., Price, B., Boyle, M.R., and Fröhlich, F. (2017). Topographical characterization of brain responses to auditory rhythmic stimuli during wakefulness and NREM sleep. *NeuroImage* 169, 57-68
- Sellers, K.K., Yu, C.X., Zhou, Z.C., Stitt, I., Li, Y.H., Radtke-Schuller, S., **Alagapan, S.**, and Fröhlich, F. (2016). Oscillatory Dynamics in the Frontoparietal Attention Network during Sustained Attention in the Ferret. *Cell Reports* 16, 2864-2874.
- Lustenberger, C., Boyle, M.R., **Alagapan, S.**, Mellin, J.M., Vaughn, B.V., and Fröhlich, F. (2016). Feedback-Controlled Transcranial Alternating Current Stimulation Reveals a Functional Role of Sleep Spindles in Motor Memory Consolidation. *Current Biology* 26, 2127-2136.
- DeMarse, T.B., Pan, L., **Alagapan, S.**, Brewer, G., and Wheeler, B.C. (2016). Feed-Forward Propagation of Temporal and Rate Information between Cortical Populations during Coherent Activation in Engineered In Vitro Networks. *Frontiers in Neural Circuits* 10.
- Alagapan, S.**, Schmidt, S.L., Lefebvre, J., Hadar, E., Shin, H.W., and Fröhlich, F. (2016). Modulation of Cortical Oscillations by Low-Frequency Direct Cortical Stimulation Is State-Dependent. *PLoS Biology* 14, e1002424.
- Alagapan, S.**, Franca, E., Pan, L., Leondopulos, S., Wheeler, B.C., and DeMarse, T.B. (2016). Structure, Function, and Propagation of Information across Living Two, Four, and Eight Node Degree Topologies. *Frontiers in Bioengineering and Biotechnology* 4, 15.
- Franca, E., Jao, P.F., Fang, S.P., **Alagapan, S.**, Pan, L., Yoon, J.H., Yoon, Y.K., and Wheeler, B.C. (2016). Scale of Carbon Nanomaterials Affects Neural Outgrowth and Adhesion. *IEEE Transactions in Nanobioscience* 15, 11-18.

- Pan, L.*, **Alagapan, S.***, Franca, E., Leondopulos, S.S., DeMarse, T.B.*, Brewer, G.J., and Wheeler, B.C. (2015). An in vitro method to manipulate the direction and functional strength between neural populations. *Frontiers in Neural Circuits* 9, 32.
- Pan, L., **Alagapan, S.**, Franca, E., DeMarse, T., Brewer, G.J., and Wheeler, B.C. (2014). Large extracellular spikes recordable from axons in microtunnels. *IEEE Transactions in Neural Systems and Rehabilitation Engineering* 22, 453-459.
- Brewer, G.J., Boehler, M.D., Leondopulos, S., Pan, L., **Alagapan, S.**, DeMarse, T.B., and Wheeler, B.C. (2013). Toward a self-wired active reconstruction of the hippocampal trisynaptic loop: DG-CA3. *Frontiers in Neural Circuits* 7, 165.
- Pan, L., **Alagapan, S.**, Franca, E., Brewer, G.J., and Wheeler, B.C. (2011). Propagation of action potential activity in a predefined microtunnel neural network. *Journal of Neural Engineering* 8, 046031.

Book Chapters

- Fröhlich, F., **Alagapan, S.***, Boyle, M.R.*, Hamilton, F.*, Li, G.*, Lustenberger, C.*, and Schmidt, S.L.* (2016). Target Engagement with Transcranial Current Stimulation. In *Transcranial Direct Current Stimulation in Neuropsychiatric Disorders: Clinical Principles and Management*, A. Brunoni, M. Nitsche and C. Loo, eds., pp. 197-222.

* Contributed equally † Mentee

TALKS	6 th Bioelectronic Medicine Summit, New York, NY	2025
	Neurology Grand Rounds, Emory University, Atlanta, GA	2024
	O'Donnell Brain Institute, UT Southwestern, Dallas, TX	2024
	Psychology Dept Seminar, Emory University, Atlanta, GA	2023
	Society for Neuroscience Annual Meeting (Nanosymposium), Washington DC	2023
	Society for Neuroscience Annual Meeting (Nanosymposium), San Diego, CA	2022
	Society of Biological Psychiatry Annual Meeting (Symposium), New Orleans, LA	2022
	Transcranial Electric Stimulation Workshop (Carolina Neurostimulation Conference), Chapel Hill, NC	2018
	Society for Industrial and Applied Mathematics Southeast Atlantic Section Conference, Chapel Hill, NC	2018
	Neuroscience Center Seminar Series, UNC Chapel Hill, NC	2017
TEACHING & MENTORING	Teaching	
	University of North Carolina at Chapel Hill	
	Guest Lecturer, Network Neuroscience	2015, 2017

University of Florida

Graduate Teaching Assistant, Neural Engineering	2011
Graduate Teaching Assistant, Neurodynamics	2010

Mentoring

Georgia Institute of Technology

Elif Ceren Fitoz, Graduate student	2022 – Present
Yenho Chen, Graduate Student	2022 – Present
Nayanika Biswas, Graduate student	2022
Tyler Albarran, Graduate student	2023 – Present
Camilla May, Research Technician	2023 – Present
Parisa Sarikhani, Postdoctoral Fellow	2024 – Present
Vivek Anand, Graduate student	2024 – Present

University of North Carolina at Chapel Hill

Mentored undergraduate and graduate students in brain stimulation studies and data analysis. Sat on 1 undergraduate honors thesis committee

Jhana Parikh, Undergraduate research assistant (Honors Thesis Committee)	2018
Trevor Mcpherson, Undergraduate research assistant	2017 – 2018
Florian Schertenleib, Visiting graduate researcher	2016 – 2017
Matt Mattoni, Undergraduate research assistant	2016 – 2017

SERVICE	Review Panelist, National Science Foundation	2025
	Nanosymposium Chair, Society for Neuroscience Annual Meeting	2022
	Conference Program Committee - Carolina Neurostimulation Conference	2018

Ad-hoc Reviewer

2016 –

Nature, Nature Communications, Biological Psychiatry, Psychological Medicine, Scientific Reports, Psychiatry Research, Neuroscience Letters, Journal of Neuroscience Methods, NeuroImage, Frontiers in Neuroscience, IEEE Journal of Biomedical and Health Informatics